Nuclear physics

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

1 · Part 1

Core · The atom · [1]

Answer: B

Explanation

An atom has a small positively charged nucleus with negatively charged electrons orbiting it. A and C give the nucleus and the electrons the wrong charges; D describes a model with no nucleus, which the nuclear model replaced.

2 · Part 1

Extended · The atom · [1]

Answer: D

Explanation

If most α-particles pass straight through, they meet nothing that deflects them, so most of the atom is empty space. A and B are supported by the few α-particles that are deflected through large angles, not by those that pass straight through. C is not something α-scattering shows.

3 · Part 1

Core · The nucleus · [1]

Answer: A

Worked solution

In ^A_Z X, the top number A is the nucleon number (protons + neutrons) and the bottom number Z is the proton number. Neutrons = A − Z = 59 − 27 = 32.

4 · Part 1

Core · The nucleus · [1]

Answer: C

Explanation

Isotopes of an element have the same number of protons (same proton number) but different numbers of neutrons. A, B and D all differ between isotopes because the neutron number differs.

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5 · Part 1 Extended The nucleus [1]

Answer: B

Explanation

Fusion is the joining of two light nuclei into a heavier nucleus; the total mass of the products is less than the total mass before, and energy is released. A describes fission. C has the mass change the wrong way round. D is wrong because mass decreases and energy is released, not absorbed.

6 · Part 1 Extended The nucleus [1]

Answer: D

Explanation

The relative charge of a nucleus equals its proton number (18 for argon, 20 for calcium), so the calcium nucleus has the greater charge. The relative mass depends on the nucleon number, which is 40 for both, so the masses are approximately the same. A and C wrongly treat the charges as equal; B has the charge comparison the wrong way round.

7 · Part 1 Core Detection of radioactivity [1]

Answer: A

Explanation

Radio waves are not ionising nuclear radiation and are not a source of background radiation. Radon gas, rocks and buildings, and cosmic rays (B, C and D) are all significant sources of background radiation.

8 · Part 1 Core Detection of radioactivity [1]

Answer: C

Worked solution

Count rate = number of counts / time. Convert the time to seconds: 4.0 × 60 = 240 s. Count rate = 1440 / 240 = 6.0 counts / s. Check: 6.0 counts / s × 60 = 360 counts / minute, and 360 × 4.0 = 1440 counts.

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9 · Part 1 Core The three types of nuclear emission [1]

Answer: A

Explanation

β-particles are fast-moving electrons; they are less ionising than α-particles but more ionising than γ-radiation, and they are stopped by a few millimetres of aluminium. B describes the ionising effect and penetration of α-particles; C gives the nature of α-particles; D describes γ-radiation.

10 · Part 1 Extended The three types of nuclear emission [1]

Answer: B

Explanation

α-particles are positive, so they are attracted towards the negative plate; β-particles are negative, so they are attracted towards the positive plate; γ-radiation is uncharged and is not deflected. β-particles have a much smaller mass than α-particles, so they are deflected more. A has the sizes of deflection the wrong way round, C has both directions reversed, and D wrongly deflects γ-radiation and not β-particles.

11 · Part 1 Core Radioactive decay [1]

Answer: D

Explanation

α-decay and β-decay both change the number of protons in the nucleus, so the nucleus becomes a different element. γ-emission does not change the number of protons or neutrons, so A and C are wrong; B leaves out β-decay, which also changes the element.

12 · Part 1 Extended Radioactive decay [1]

Answer: C

Worked solution

α-decay: proton number − 2, nucleon number − 4. First β-decay: proton number + 1, nucleon number + 0. Second β-decay: proton number + 1, nucleon number + 0. Total: proton number − 2 + 1 + 1 = 0 (the same); nucleon number − 4. The final nucleus is an isotope of the original element.

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13 · Part 1 Core Half-life [1]

Answer: B

Worked solution

Number of half-lives = 24 / 8.0 = 3. After 8 h: 720 / 2 = 360; after 16 h: 360 / 2 = 180; after 24 h: 180 / 2 = 90 counts / minute. Check: 90 × 2 × 2 × 2 = 720.

14 · Part 1 Core Safety precautions [1]

Answer: A

Explanation

Ionising radiation can damage cells, causing cell death, mutations and cancer. B is a common misconception: exposure to radiation does not make cells radioactive. C is wrong because radiation from outside the body can also harm cells. D is wrong because even small doses can cause mutations.

15 · Part 1 Extended Safety precautions [1]

Answer: C

Explanation

Tongs keep the source further away from the body, and increasing the distance from the source reduces the amount of radiation reaching the body. A is wrong because tongs are not a shield and do not absorb all the radiation. B and D are wrong because nothing done to the source changes its rate of decay or its half-life.

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16 · Part 1 Core The atom [5]

Mark scheme

(a) **[2]**

- (small) positively charged nucleus (at the centre) (1)
- negatively charged electrons orbit (around) the nucleus (1)

(b) **[1]**

- 11 (1)

(c) **[2]**

- positive ion (1)
- it now has more protons than electrons / more positive charges than negative charges (1)

ALLOW it has 11 protons but only 10 electrons

REJECT it gains a proton / loses a negative proton

Worked solution

(a) An atom has a tiny nucleus at its centre, which is positively charged. Negatively charged electrons move in orbits around the nucleus.

(b) A neutral atom has no overall charge, so the number of electrons (each −1) must equal the number of protons (each +1). So there are 11 electrons.

(c) Losing an electron removes one negative charge. The atom now has 11 protons (+11) and 10 electrons (−10), so its overall charge is +1. Atoms that lose electrons form positive ions.

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17 · Part 1 Core The nucleus [7]

Mark scheme

(a) **[1]**

- the number of protons in the nucleus (of an atom) (1)

(b) **[1]**

- the (total) number of protons and neutrons in the nucleus / the number of nucleons in the nucleus (1)

(c) **[3]**

- protons: 29 (1)
- neutrons: 36 (1)
- electrons: 29 (1)

(d) **[2]**

- they have the same number of protons / same proton number (1)
- they have different numbers of neutrons / different nucleon numbers (1)

Worked solution

(a) and (b) The proton number Z is the number of protons in the nucleus. The nucleon number A is the total number of protons and neutrons (nucleons) in the nucleus.

(c) In ^{65}_29 Cu, Z = 29, so there are 29 protons. Neutrons = A − Z = 65 − 29 = 36. A neutral atom has as many electrons as protons, so there are 29 electrons.

(d) Both nuclei have 29 protons, which makes them the same element (copper). ^{63}_29 Cu has 63 − 29 = 34 neutrons and ^{65}_29 Cu has 36 neutrons, so they differ only in the number of neutrons. Atoms of the same element with different numbers of neutrons are isotopes.

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18 · Part 1 Extended The nucleus [6]

Mark scheme

(a) **[1]**

- (nuclear) fission (1)

(b) **[1]**

- 38 (1)

(c) **[2]**

- total nucleon number before = 1 + 235 = 236 / 140 + 94 + x = 236 (1)
- x = 2 (1)

(d) **[2]**

- the total mass after the reaction is less than the total mass before (1)
- energy is released (1)

IGNORE any attempt to give a value of the mass or energy change

Worked solution

(a) A heavy uranium nucleus splits into two smaller nuclei after absorbing a neutron. Splitting a nucleus is nuclear fission.

(b) Proton numbers must balance: 0 + 92 = 54 + Z + 0, so Z = 92 − 54 = 38.

(c) Nucleon numbers must balance. Before: 1 + 235 = 236. After: 140 + 94 + x(1) = 234 + x. So 234 + x = 236 and x = 2. Two neutrons are released. Check the proton numbers with x = 2: 54 + 38 + 2 × 0 = 92, which matches.

(d) In fission the total mass of the products is slightly less than the total mass of the particles before the reaction. This mass is not lost for nothing: energy is released, mostly as kinetic energy of the products.

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19 · Part 1 Extended The nucleus [4]

Mark scheme

(a) **[1]**

- the relative charge (on the nucleus) is equal to the proton number / relative charge = +Z (1)

ALLOW charge is (directly) proportional to proton number

(b) **[1]**

- +26 (1)

ALLOW 26

(c) **[2]**

- relative mass depends on / is (approximately) proportional to the nucleon number, 56 compared with 14 (1)
- 4 (times greater) (1)

REJECT answers that compare proton numbers (26 / 7, about 3.7)

Worked solution

(a) Each proton has a relative charge of +1 and neutrons are uncharged, so the relative charge on a nucleus is equal to its proton number.

(b) ^{56}_26 Fe has 26 protons, so the relative charge is +26.

(c) Protons and neutrons have almost the same mass, so the relative mass of a nucleus is set by the nucleon number, not the proton number. Iron has 56 nucleons and nitrogen has 14, so the iron nucleus has 56 / 14 = 4 times the relative mass of the nitrogen nucleus. Check: 14 × 4 = 56.

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20 · Part 1 Core Detection of radioactivity [4]

Mark scheme

(a) **[1]**

- (ionising) radiation that is always present in the environment / around us, from natural (and artificial) sources (1)

ALLOW radiation detected when no (radioactive) source is present, if it refers to the surroundings

(b) **[3]**

- any three from (1) each:
- radon gas (in the air)
- rocks / buildings / building materials / the ground
- food and drink
- cosmic rays / radiation from space / the Sun

REJECT mobile phones, microwaves, radio waves, Wi-Fi, electricity cables

Worked solution

(a) Background radiation is the low level of ionising radiation that is present all the time, everywhere, from sources in our surroundings. A detector picks it up even when no source is nearby.

(b) The main sources are radon gas, which seeps out of the ground into the air; rocks and building materials such as granite and brick; food and drink, which contain small amounts of radioactive isotopes; and cosmic rays arriving from space. Radio waves and microwaves are not ionising nuclear radiation and are not part of background radiation.

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21 · Part 1 Core Detection of radioactivity [5]

Mark scheme

(a) **[2]**

- count rate = 270 / 6.0 (1)
- 45 counts / minute (1)

(b) **[1]**

- 33 counts / minute (1)

ALLOW 33.0 counts / minute

(c) **[1]**

- radon gas collects / builds up in the cellar / the cellar is surrounded by rocks (and soil) that contain radioactive materials (1)

ALLOW less ventilation, so radon is not removed

REJECT answers based on electrical equipment or mobile phone signals

(d) **[1]**

- radioactive decay / emission is random (so the number of counts in a given time varies) (1)

ALLOW spontaneous and random

Worked solution

(a) Count rate = number of counts / time = 270 / 6.0 = 45 counts / minute.

(b) Count rate = 330 / 10.0 = 33 counts / minute. Check: 33 × 10.0 = 330.

(c) Radon is a radioactive gas that comes out of the ground. It can build up in rooms below ground level, especially if there is little air flow. The cellar walls are also surrounded by rock and soil, which contain radioactive materials.

(d) The decay of nuclei is random, so it is impossible to predict exactly how many decays will be detected in any time interval. Repeating the measurement gives a slightly different number even though nothing has changed.

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22 · Part 1 Extended Detection of radioactivity [6]

Mark scheme

(a) **[1]**

- 150 / 5.0 = 30 counts / minute (1)

(b) **[3]**

- measured count rate = 2280 / 4.0 = 570 counts / minute (1)
- subtracts the background count rate: 570 − 30 (1)
- 540 counts / minute (1)

ECF from the value in (a)

(c) **[1]**

- 45 − 30 = 15 counts / minute (1)

ECF from the value in (a)

(d) **[1]**

- the background count rate is a large fraction / proportion of the measured count rate (so ignoring it gives a large error) (1)

ALLOW background is two-thirds of the measured count rate / background is twice the count rate from the source

Worked solution

(a) Background count rate = 150 / 5.0 = 30 counts / minute.

(b) First find the measured count rate with the source present: 2280 / 4.0 = 570 counts / minute. This includes the background. The count rate due to the source alone (the corrected count rate) = 570 − 30 = 540 counts / minute. Check: (540 + 30) × 4.0 = 2280.

(c) Corrected count rate = 45 − 30 = 15 counts / minute.

(d) Near the source, the background (30) is small compared with 570, so ignoring it changes the answer only a little. Far away, the background (30) is two-thirds of the measured value (45). Without the correction the source's count rate would be taken as 45 instead of 15, which is three times too large.

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23 · Part 1 Core The three types of nuclear emission [5]

Mark scheme

(a) **[3]**

- no α-particles: the paper makes (almost) no difference to the count rate (1)
- β-particles: the 3 mm of aluminium greatly reduces the count rate (1)
- γ-radiation: the count rate does not fall to zero with aluminium / the count rate hardly changes when the aluminium is doubled to 6 mm (1)

ALLOW small changes such as 820 to 816 or 210 to 208 described as random variation

(b) **[1]**

- electromagnetic radiation / electromagnetic waves (1)

REJECT particles / electrons / helium nuclei

(c) **[1]**

- the radiation is emitted in any direction / its direction cannot be predicted (1)

Worked solution

(a) α-particles are stopped by a sheet of paper. The count rate only changes from 820 to 816, which is a tiny change caused by the random nature of decay, so the source does not emit α-particles. β-particles are stopped by a few millimetres of aluminium. The count rate drops sharply from 816 to 210 with 3 mm of aluminium, so the source emits β-particles. γ-radiation passes through aluminium easily. Radiation is still detected through the aluminium, and doubling the thickness to 6 mm barely changes the count rate, so what is left is γ-radiation. The source emits β-particles and γ-radiation.

(b) γ-radiation is electromagnetic radiation (a wave), not a particle.

(c) Random in direction means a nucleus can emit its radiation in any direction, and it is not possible to predict which direction it will be.

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24 · Part 1 Extended The three types of nuclear emission [7]

Mark scheme

(a) **[2]**

- α-particles have a large (relative) charge of +2, so they exert large (electric) forces on electrons in atoms / pull electrons out of atoms (1)
- α-particles have a large kinetic energy (because of their large mass), so they can ionise many atoms (giving up some energy each time) (1)

ALLOW α-particles are massive and slow, so they spend longer near each atom

(b)(i) **[2]**

- downwards / towards the bottom of the page (1)
- α-particles are positively charged, the opposite charge to β-particles (1)

(b)(ii) **[2]**

- the deflection of the α-particles is (much) smaller (1)
- α-particles have a much greater mass (than β-particles) (1)

ALLOW α-particles have much greater momentum

(c) **[1]**

- not deflected because γ-radiation has no charge (1)

REJECT not deflected because it is too fast

Worked solution

(a) Ionisation happens when radiation removes electrons from atoms. An α-particle has a charge of +2, the largest of the three types, so it attracts electrons in the atoms it passes very strongly. It also has a large kinetic energy because it has a large mass. Each ionisation uses some of this energy, so an α-particle ionises a large number of atoms in a very short distance before it stops.

(b)(i) A magnetic field pushes a moving positive charge in the opposite direction to a moving negative charge travelling the same way. β-particles are negative and go upwards, so α-particles, which are positive, are deflected downwards.

(b)(ii) An α-particle has thousands of times the mass of a β-particle, so it is much harder to change its direction. Its deflection is much smaller.

(c) γ-radiation is an uncharged electromagnetic wave, so the magnetic field exerts no force on it and it continues in a straight line.

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25 · Part 1 Core Radioactive decay [5]

Mark scheme

(a) **[2]**

- a (random) change in an unstable nucleus (1)
- in which α-particles, β-particles and / or γ-radiation are emitted (1)

ALLOW nuclear radiation is emitted, for the second mark

(b) **[2]**

- a different element (1)
- an α-particle contains protons, so the number of protons in the nucleus changes / decreases (1)

ALLOW the proton number changes (by 2)

(c) **[1]**

- radioactive decay is random / spontaneous (so the time at which any one nucleus decays cannot be predicted) (1)

Worked solution

(a) Radioactive decay is a change in an unstable nucleus. During the change the nucleus emits α-particles or β-particles, and / or γ-radiation.

(b) An α-particle is made of two protons and two neutrons. When a nucleus emits one, it loses two protons. The element is fixed by the number of protons, so the new nucleus belongs to a different element.

(c) Decay is spontaneous and random. Nothing outside the nucleus triggers it, and every unstable nucleus has the same chance of decaying in a given time, so there is no way of telling which one will decay next.

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26 · Part 1 Extended Radioactive decay [5]

Mark scheme

(a) **[1]**

- a neutron changes into a proton and an electron / neutron → proton + electron (1)

REJECT a proton changes into a neutron

(b) **[2]**

- A = 32 (1)
- Z = 16 (1)

(c) **[1]**

- neither the proton number nor the nucleon number changes (1)

(d) **[1]**

- a neutron changes into a proton, so the number of excess neutrons is reduced (1)

Worked solution

(a) In β-emission a neutron in the nucleus changes into a proton and an electron. The electron is emitted as the β-particle; the proton stays in the nucleus.

(b) Nucleon numbers balance: 32 = A + 0, so A = 32. Proton numbers balance: 15 = Z + (−1), so Z = 16. Check: the proton number goes up by 1 because a neutron became a proton, and 15 + 1 = 16.

(c) γ-radiation is electromagnetic radiation with no charge and no mass, so emitting it removes energy but does not change the number of protons or neutrons.

(d) $^{32}_{15}$P has 15 protons and 17 neutrons, which is too many neutrons for a stable nucleus. β-emission turns one neutron into a proton, giving 16 protons and 16 neutrons. The excess of neutrons is reduced, so the nucleus is more stable.

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27 · Part 1 Extended Radioactive decay [6]

Mark scheme

(a) **[1]**

- the nucleus is too heavy / has too many nucleons (1)

ALLOW it has too many neutrons / an excess of neutrons

(b) **[3]**

- α-particle written as $^{4}_{2}\alpha$ or $^{4}_{2}\text{He}$ (1)
- nucleon number of radon 222 (1)
- proton number of radon 86 (1)

(c) **[2]**

- nucleon number 218 (1)
- proton number 84 (1)

ECF from the radon nuclide given in (b) (nucleon number 4 less, proton number 2 less)

Worked solution

(a) Radium-226 has a very large nucleus with 226 nucleons. Nuclei this heavy are unstable, and they can become more stable by emitting α-particles.

(b) An α-particle is a helium nucleus, $^{4}_{2}\alpha$. Nucleon numbers balance: 226 = A + 4, so A = 222. Proton numbers balance: 88 = Z + 2, so Z = 86. The equation is $^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\alpha$.

(c) A second α-decay again removes 4 nucleons and 2 protons: nucleon number 222 − 4 = 218 and proton number 86 − 2 = 84. The polonium nucleus is $^{218}_{84}\text{Po}$. Check: 218 + 4 + 4 = 226 and 84 + 2 + 2 = 88.

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28 · Part 1 Core Half-life [6]

Mark scheme

(a) **[2]**

- the time taken for half the nuclei (of that isotope) (1)
- in any sample to decay (1)

REJECT half the time for the sample to decay / the time for half the mass of the sample to disappear

(b) **[2]**

- the count rate falls to half its value: e.g. from 960 to 480 / from 480 to 240 / from 680 to 340 (1)
- 4.0 hours (1)

(c) **[2]**

- 16.0 hours is 4 half-lives / count rate at 8.0 hours (240) halves twice more (1)
- 60 counts per minute (1)

ECF from the half-life in (b)

Worked solution

(a) The half-life of an isotope is the time taken for half the nuclei of that isotope in any sample to decay.

(b) Pick any value and find when it has halved. At 0 hours the count rate is 960; it is 480 at 4.0 hours. So the half-life is 4.0 hours. Check with another pair: 680 at 2.0 hours falls to 340 at 6.0 hours, again 4.0 hours.

(c) 16.0 hours is 16.0 / 4.0 = 4 half-lives. 960 → 480 → 240 → 120 → 60, so the count rate is 60 counts per minute. Check: $60 \times 2^4 = 60 \times 16 = 960$.

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29 · Part 1 Extended Half-life [7]

Mark scheme

(a) **[1]**

- 700 − 60 = 640 counts / minute (1)

(b) **[3]**

- subtracts the background from the readings, e.g. 380 − 60 = 320 (1)
- the corrected count rate halves from 640 to 320 / from 320 to 160 (1)
- 10 minutes (1)

REJECT a half-life of about 11–12 minutes found by halving the uncorrected readings

(c) **[3]**

- X (1)
- β-particles are partly absorbed by paper, so the count rate changes when the thickness changes / α-particles would be completely stopped by the paper and γ-radiation would pass through with almost no change (1)
- the long half-life means the count rate from the source stays (almost) constant over time / the source does not need replacing often (1)

REJECT Z (Z's count rate would fall too quickly)

Worked solution

(a) The count rate due to the sample = measured − background = 700 − 60 = 640 counts / minute.

(b) Subtract 60 from every reading to get the corrected count rates: 640, 453, 320, 226, 160, 113 counts / minute. The corrected rate halves from 640 at 0 minutes to 320 at 10.0 minutes, and from 320 to 160 between 10.0 and 20.0 minutes, so the half-life is 10 minutes. If you halve the uncorrected readings (700 → 350) you get the wrong answer, because the background does not decay.

(c) To monitor paper thickness the radiation must be partly absorbed by the paper, so that a thicker sheet gives a noticeably lower count rate. α-particles would be completely stopped by paper, and γ-radiation would pass straight through almost unaffected, so a β-emitter is needed: X or Z. Z has a half-life of only 15 minutes, so its count rate would drop quickly even with no change in thickness, and it would need replacing constantly. X has a half-life of 28 years, so its count rate stays almost constant and any change in the reading is caused by a change in paper thickness. X is the best choice.

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30 · Part 1 Core Safety precautions [6]

Mark scheme

(a) **[2]**

- any two from (1) each:
- cell death / kills cells
- mutation(s)
- cancer

ALLOW radiation burns / radiation sickness

REJECT makes the person radioactive

(b) **[2]**

- in a lead-lined box / container (1)
- kept in a locked cupboard / secure store, (labelled with a radiation warning sign,) away from people (1)

(c) **[2]**

- any two from (1) each:
- handle the source with (long-handled) tongs / never touch it with the hands
- hold the source at arm's length / keep it as far from the body as possible
- point the source away from people / never point it at anyone
- keep the source out of its box for as short a time as possible / return it to its box straight after use

ALLOW carry it in its lead-lined box when moving it

Worked solution

(a) Ionising radiation damages the cells of living things. It can kill cells, it can cause mutations (changes in the DNA), and these mutations can lead to cancer.

(b) The sources are kept in lead-lined boxes, because lead absorbs the radiation. The boxes are kept in a locked cupboard with a warning sign, so that only trained staff can reach them.

(c) The teacher should carry the source in its lead-lined box, lift it out with long tongs rather than fingers, keep it at arm's length and pointed away from everyone, and put it back in its box as soon as the experiment is finished. These steps keep the source further away from people and reduce the time anyone is exposed.

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